

Aswin Neelakandan

Curriculum vitae (17th Sep 2026)

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🌐 [Aswin Neelakandan](#)

🔗 [Aswin-Neel](#)

EDUCATION

Indian Institute of Science Education and Research (IISER) Berhampur

Expected May 2027

BS-MS, Physics

Berhampur, Odisha

- **Relevant Coursework:** Mathematical methods, Mechanics, Electromagnetism, Calculus of one variable, Waves and introductory optics, Linear algebra, Quantum physics, Multivariable calculus, Technical writing, Classical mechanics, Quantum mechanics - I,II, Thermal physics, Numerical methods and analysis (computational), Statistical Mechanics, Electrodynamics, Special Theory of Relativity, Condensed Matter Physics, Group Theory, Atomic and Molecular Physics, Nuclear and Particle Physics, General Theory of Relativity.

RESEARCH INTERESTS

I am broadly interested in the multi-wavelength studies of galaxies, galaxy clusters and large-scale structure, using statistical and computational methods to probe the dynamics of compact objects, feedback processes, and structure formation.

RESEARCH EXPERIENCE

Statistical Analysis of X-ray Cavities using LoTSS DR3 and Chandra

May 2025 - Present

PI: Prof. Huub Rottgering, Leiden University

co-PI: Prof. Abhirup Datta, IIT-Indore

Indore, Madhya Pradesh

- * Built batch pipeline detecting AGN feedback signatures (X-ray cavities) across 500 clusters processing from raw evt files to Sherpa/CIAO beta-model fitting
- * Probing AGN feedback at a survey scale by computing cavity energetics and comparing with mechanical radio power
- * uGMRT Cycle 51 proposal accepted (Band-3) for observations of galaxy-cluster cavity systems — synchrotron ageing, spectral-index mapping, equipartition B-field estimates.

QSO's as tracers of our Universe

Jun 2024 - Present

PI: Dr. Shadab Alam, TIFR Mumbai

Mumbai, Maharashtra

- * Worked on arriving at the projected correlation function to validate theory with respect to observed data from surveys like eBOSS and eROSITA.
- * Working on a relation between LSS traced by QSO's and the underlying dark matter halo mass distribution.
- * Computed BH number density, merger mass-ratio distribution $P(q)$, and merger rate density $R(q)$ with beta-distribution fits, to be further correlated with kick velocities in LSS.

TALKS/PRESENTATIONS

MCMC as a Statistical Method

15/09/25

Journal Club Talk

Berhampur, Odisha

- Presented Markov Chain Monte Carlo as a method for parameter estimation and uncertainty quantification, with applications to astrophysical model fitting using emcee.

Introduction to Radio Astronomy

23/08/25

Invited Talk, Kendriya Vidyalaya (KV)

Berhampur, Odisha

- Introduced high schoolers to the fundamentals of radio astronomy, aiming to build early interest in observational techniques.

COURSES AND PROJECTS

- Supervised Machine Learning: Regression and Classification | *Stanford Online*
- From big bang to dark energy | *University of Tokyo*
- Holographic Principle - Report
- Developed ML based methods to classify AGN morphologies: NATs/WATs and estimate radio lobe volume.

LEADERSHIP AND OUTREACH

TEDx

May 2024 - Sep 2024

Core member + Tech lead

Berhampur, Odisha

- Lead the tech team, worked on integration with the main fest's website.
- Coordinated with tedx speakers, gained insights with regards to their expertise in their various fields.

Episteme - science magazine

May 2024 - Dec 2025

Core member + Tech lead

Berhampur, Odisha

- Managing, structuring, publishing and proof reading science communication material to be added to the magazine
- Remade the website, after a catastrophic loss of the prior one, due to hosting service migration problems.
- Handling maintenance, upkeep and updates to the website as well.

Stream - Outreach

Jan 2024 - Feb 2024

Member of Organizing committee

Berhampur, Odisha

- Was an outreach event aimed at developing a scientific temperament in young school students.
- Created multiple engaging presentations and experiments, that were showcased to the students, to foster their growth and interest in the fields of astronomy, physics and programming.
- Arranged food, and other basic necessities for the visiting students.

Naxatra - Astronomy club

Dec 2023 - Present

Core member

Berhampur, Odisha

- Organized events aimed at increasing interest in, and awareness about astronomy, in a holistic and engaging manner.
- Organized telescope sessions, especially during celestial events, aimed at teaching the population of the university about the inner workings of a telescope and the skills necessary to operate one.
- Hosted multiple online and offline talks presented by both students and professors.

Lovelace society - Coding and robotics club

Sep 2023 - Jun 2025

Coordinator

Berhampur, Odisha

- Organized events and workshops to test and improve the coding skills of the participants.

137 Inverse - Physics club

Sep 2023 - Apr 2025

Core member

Berhampur, Odisha

- Organized educational and fun events to foster a scientific approach to thinking amongst college students, like, "Pandora's box" and "2024: A Space Odyssey"
- Managed the club-run physics newsletter, "Matter matters", to ensure routine and timely publishing, along with quality and reach of articles.
- Hosted, arranged and organized multiple online and offline talks presented by both professors and students.

Covid care teens - NGO

May 2021 - Dec 2021

Core member

Chennai, Tamil Nadu

- Organized and created easy access mechanisms for the large amounts of data regarding the available government and private health services during the pandemic.
- Participated in upkeep of website and management of communication channels.

SKILLS

Programming: $\text{\LaTeX}$, Python, HTML, CSS, JS, Jupyter, Git, Linux

Other: Microsoft Office, Adobe Creative Suite